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Synthesis and Oxygen Evolution Reaction Application of a Co–Cd Based Bimetallic Metal-Organic Framework

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In the realm of renewable energy technologies, the development of efficient and durable electrocatalysts is paramount, especially for applications like electrochemical water splitting. This research focuses on synthesizing a novel bimetallic metal-organic framework (BMMOF11) using earth-abundant elements, cobalt (Co) and cadmium (Cd). BMMOF11 showcases a distinctive structure with distorted octahedral chains of CoO and CdO, linked by benzene tricarboxylic acid (BTC). Our study primarily investigates the electrocatalytic efficiency of BMMOF11, particularly in water oxidation reactions. For practical analysis, BMMOF11 was anchored onto nickel foam, forming BMMOF11/NF, to evaluate its electrocatalytic properties. Electrochemical testing revealed that BMMOF11/NF begins water oxidation at an onset potential of 1.62 V versus RHE, demonstrating high activity with a lower overpotential of 0.4 V to achieve a current density of 10 mA/cm^2. Moreover, BMMOF11/NF maintained stable water splitting performance, sustaining a current density of approximately 70 mA/cm^2 under a voltage of 1.9 V relative to RHE. These findings indicate that BMMOF11/NF is a promising candidate for large-scale electrochemical water splitting, offering a blend of high activity and stability.

1. Introduction

In light of the increasing global energy crises and environmental concerns, it is essential to develop renewable and sustainable energy sources to replace nonrenewable sources like fossil fuels. As an energy carrier with a very high energy density (approximately 142 MJ/kg) and zero carbon content, hydrogen has been considered a promising carbon-free fuel.[1,2] In regard to the prevailing conditions, electrochemical water splitting emerges as a feasible and efficient approach for mitigating acute energy shortages and reducing greenhouse gas emissions. This process involves passing the electrical current through the water to produce O_2 and H_2, which can subsequently be stored and used a an energy source.[3–8] The hydrogen evolution reaction (HER) and the oxygen evolution reaction (OER) are recognized as crucial half-reactions in the water-splitting process, taking place on the cathode and anode, respectively, within a water electrolyzer.[9] However, mass production of H_2 is limited by the relatively large overpotentials to drive OER and HER processes. Commercial electrolyzers currently operate at applied potentials of 1.8–2.2 V, which is approximately 50% greater than the thermodynamically required potential of 1.23 V at ambient temperature. Therefore, significant efforts have been made to develop active electrocatalysts to decrease the overpotential and increase the energy efficiency of the process.[10–12] However, the majority of developed catalysts fail to remain highly efficient and stable over the wide range of pH and operate at higher overpotentials. Thus, developing bimetallic electrocatalysts, especially based on metal-organic frameworks (MOFs), with increased activity toward electrochemical water splitting can streamline this process.[13–15]

In the development of efficient and durable electrocatalysts for water splitting, MOFs have emerged as a promising category due to their unique structural properties. MOFs are characterized by their porous crystalline nature, high surface area, and tunable internal porosity, which make them particularly suitable for energy conversion and storage applications.[16–19] The high surface area and abundance of molecular metal sites in MOFs are crucial for enhancing electrocatalytic processes, as they provide ample active sites for the HER and OER.[20,21] This positions MOFs as potential candidates for improving the energy efficiency of electrochemical water splitting. Unlike traditional applications where MOFs serve as precursors for the synthesis of metal-based compounds, our study explores the direct application of MOFs as electrocatalysts. This approach is relatively novel and addresses the limitations of MOFs, including their instability and inadequate conductivity, which have hindered their practical application in water splitting.[22,23] Within the realm of functional nanoporous materials, MOFs represent a relatively recent category that exhibits captivating structural characteristics and promising application prospects. MOFs consist of metal clusters and organic linkers, rendering them a distinctive and intriguing class.[24] These materials have been

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widely utilized due to their porous crystalline nature, high surface area, tunable internal porosity, and abundance of inherent molecular metal sites.[25] MOFs have wide-ranging applications in energy conversion and storage, such as batteries,[26,27] hydrogen-oxygen polymer electrolyte membrane fuel cells (PEMFC) for the oxygen reduction reaction (ORR),[28] HER, and OER.[29] Nevertheless, the practical application of MOF materials is sometimes hindered by their instability and inadequate conductivity. Consequently, in many instances, MOFs are employed as a precursor for the synthesis of metal-based compounds and porous nanocarbon materials through pyrolysis. This approach is favored due to the high surface area, abundance of active sites, and porous characteristics exhibited by MOFs.[27] However, the inherent active sites and ordered structure of MOFs can be entirely destroyed due to pyrolysis. In addition, the MOF particles can agglomerate during pyrolysis, which can reduce the electron conductivity and the transfer of reactants and products in an electrochemical reaction. The most effective water-splitting catalysts are noble metals, such as Pt-based precious metal alloys and Ir/Ru-based oxides. But these catalysts have inimical disadvantages, which include their high cost, instability, scarce earth abundance, and operational challenges arising from the usage of two different catalysts for each half-reaction. Consequently, there is an ongoing need for bifunctional electrocatalysts to show both HER and OER catalytic capabilities in the same electrolyte solutions.[30] Different types of transition metals and their oxides and hydroxides,[31] chalcogenides,[32] and phosphides[33] have been investigated as potential bifunctional catalysts for water splitting. Several MOFs-derived materials have been explored as potential electrocatalysts for water splitting.[34,35] However, very few MOFs have been directly explored as electrocatalysts for water splitting owing to their high resistance and instability.[36] So, it is extremely important to probe the direct application of MOFs as electrocatalysts for water splitting.

In the development of our novel MOF, we strategically chose commonly used elements i.e., cobalt (Co) and cadmium (Cd) for their synergistic properties when combined in a bimetallic system. The rationale behind using Co and Cd together lies in their complementary electrochemical characteristics, which are essential for enhancing the catalytic efficiency and stability in water splitting applications. Co is known for its excellent redox properties, while Cd contributes to the structural stability and electronic conductivity of the MOF. This unique combination in BMMOF11 is expected to offer improved catalytic performance compared to monometallic counterparts. It is important to note that BMMOF11 is a newly developed MOF and, to the best of our knowledge, has not been reported previously in the literature. In this study, commonly used elements (Co and Cd) are utilized to produce an efficient electrocatalyst (BMMOF11) with enhanced electrocatalytic activities. The crystal structure of BMMOF11 is characterized by a monoclinic bimetallic arrangement, wherein zig-zag chains composed of alternating and vertex-sharing CdO_6 and CoO_6 distorted octahedra are observed along the [001] direction. These chains are bridged by the benzene-1,3,5-tricarboxylate (BTC) linker, giving rise to the 3D network. The prepared BMMOF11 is immobilized on nickel foam (NF) to study its electrocatalytic properties towards water oxidation. The electrochemical analyses show that the BMMOF11/NF reaches a high current density of 124 mA/cm^2 at 1.9 V (*vs.* reversible hydrogen electrode (RHE)) and an electrochemical water oxidation starting potential of ~1.62 V *vs.* RHE. The BMMOF11/NF electrocatalyst is exceptionally active, requiring only 0.4 V of overpotential to achieve a current density of 10 mA/cm^2. Water can be continuously split into O_2 and H_2 at a current density of roughly 70 mA/cm^2 with an applied voltage of 1.9 V *vs.* RHE, as demonstrated by a long-term electrolysis stability test utilizing the BMMOF11/NF electrocatalyst. Our study indicates that BMMOF11/NF possesses promising catalytic activity for large-scale electrochemical water splitting.

2. Results and Discussion

2.1. Crystal structure of BMMOF11

The compound denoted by the chemical formula $[CoCd(BTC)(NO_3)(DMF)_2]$ exhibits crystallization in the monoclinic system, specifically inside the $P2_1/c$ space group. The asymmetric unit consists of one Co(II) ion, one Cd(II) ion, one BTC linker, one nitrate anion, and two dimethylformamide (DMF) molecules that act as ligands. The BTC linker exhibits a bidentate bridging mode $Co(\mu_2\text{-}\kappa^2\text{-}O,O)Cd$ for two of its carboxylate groups, whereas the third carboxylate adopts a chelating mode (κ^2-O,O)Cd. The nitrate ion demonstrates three different coordination modes in the compound: a chelating mode (κ^2-O,O)Co, a bidentate bridging mode $Cd(\mu_2\text{-}\kappa^2\text{-}O,O)Cd$, and a $Cd(\mu_2\text{-}\kappa^1\text{-}O)Co$ bridging mode. The Co(II) coordination sphere, $[CoO_6]$, consists of oxygen donors derived from two molecules of DMF and two carboxylate groups, in addition to the chelating nitrate (Figure S1). The Cd(II) coordination sphere $[CdO_6]$ comprises of oxygen donors from two nitrates, one chelating carboxylate, and two bridging carboxylates (Figure S2). The bond distances and bond angles observed in $[CoCd(BTC)(NO_3)(DMF)_2]$ are similar to those reported for the MOF $[Cd_2Co(BTC)_2(H_2O)_4].2H_2O$.[37] The compound consists of zig-zag chains along the [001] direction, formed by alternating and vertex sharing of $[CdO_6]$ and $[CoO_6]$ distorted octahedra. The interconnection of these chains is facilitated by the BTC linker, resulting in the formation of a 3D network (Figure 1).

2.2. Characterization

Powder X-ray diffraction (PXRD) confirmed the crystallinity and phase purity of BMMOF11 (Figure 2), showing that it is highly crystalline and single-phase. This structural characterization also corroborated the coherence in the connectivity and integrity of the framework between the simulated and the as-synthesized MOF, with the characteristic peaks at $2\theta = 8.8$, 10.3, 10.8, and 12.9. The FTIR bands (Figure S3) at 717 cm^{-1} and 771 cm^{-1} correspond to the out-of-plane aromatic C–H bending modes of the linker. The peak at 1105 cm^{-1} is assigned to the aromatic

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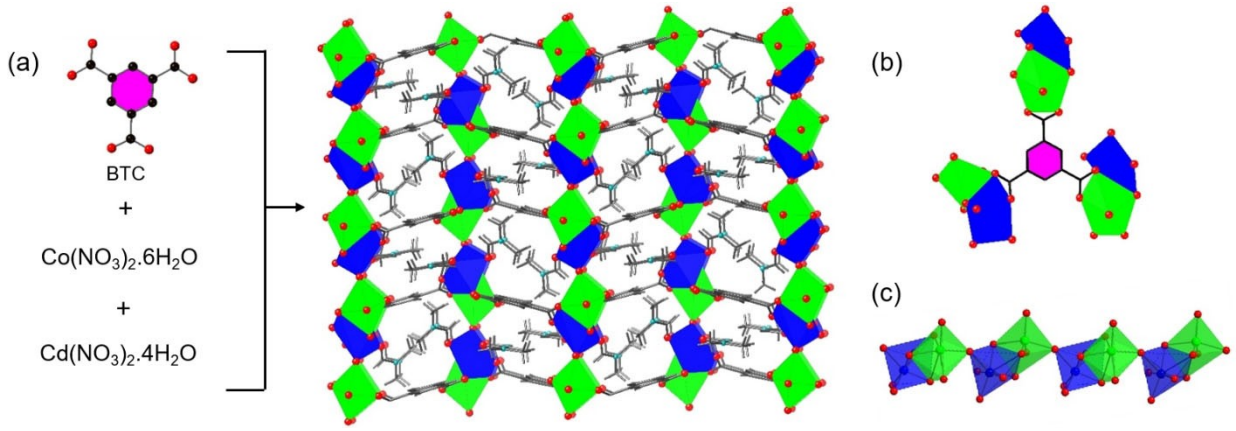


Figure 1. (a) Single-crystal structure of BMMOF11 constructed from BTC, Co^{2+}, and Cd^{2+} building blocks. (b) A single unit showing the binding of the BTC with Co^{2+} and Cd^{2+}. (c) Inorganic Secondary Building Unit, comprising Co^{2+} and Cd^{2+}. Atom colors: Cd, green polyhedral; Co, blue polyhedral; C, black; O, red. H atoms are omitted for clarity.

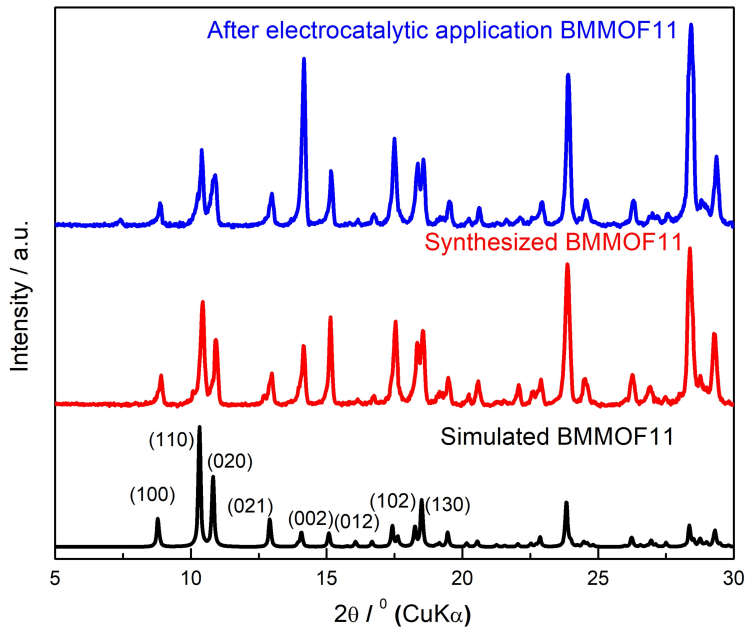


Figure 2. PXRD analysis of BMMOF11. The simulated pattern (black) of BMMOF11 matches well with the as-synthesized sample of BMMOF11 (red) and after electrocatalytic application (blue).

C–H in-plane bending, whereas the sharp peaks at 1369, 1435 cm^{-1}, 1534, and 1612 cm^{-1} are most likely attributable to the symmetric and asymmetric C–O stretching modes of the COO^- group. The N_2 adsorption isotherms of BMMOF11 at 77 k revealed a sharp surge in the uptake at relatively low pressure, referring to a microporous nature with a Type I isotherm. The Brunauer-Emmett-Teller surface area calculated was 584 m^2/g (Figure 3).

Thermogravimetric analysis (TGA) was used to study the thermal stability of BMMOF11. The TGA displayed an initial weight loss of about 3.6% in the temperature range of 150 °C to 200 °C. This decrease can be attributed to the volatilization of DMF that was entrapped within the pores. Subsequently, a sudden reduction in weight of 60.4% occurs at a temperature of 300 °C, which can be attributed to the thermal decomposition of the entire framework. The final residue, which constitutes 36.0% of the total, can be attributed to the presence of cobalt and cadmium oxides (Figure 4).

Figures 5a and 5b present digital photographs, which offer compelling visual evidence of the well-formed BMMOF11 structures. Additionally, Figure 5c displays a field emission scanning electron microscopy (FESEM) image, which effectively highlights the impressive morphology of the structures. Furthermore, Figures 5c–f depict FESEM images of BMMOF11 in its bulk form, exhibiting well-defined, highly crystalline particles with a cubic morphology. Upon closer inspection at high magnification, the FESEM images reveal a highly porous nature covered by small particles. Figures 5g–i present FESEM images of BMMOF11 after grinding with a mortar and pestle. This physical processing step significantly decreases particle size and

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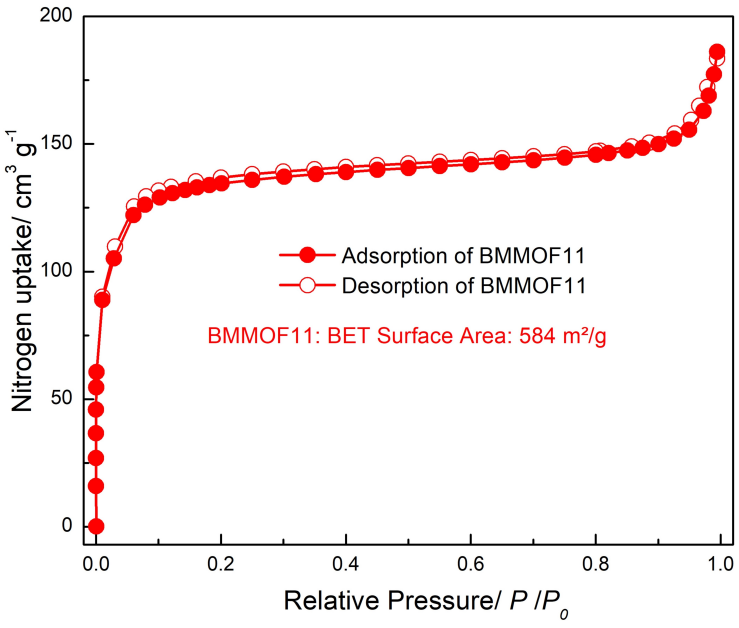


Figure 3. Nitrogen adsorption isotherm of BMMOF11.

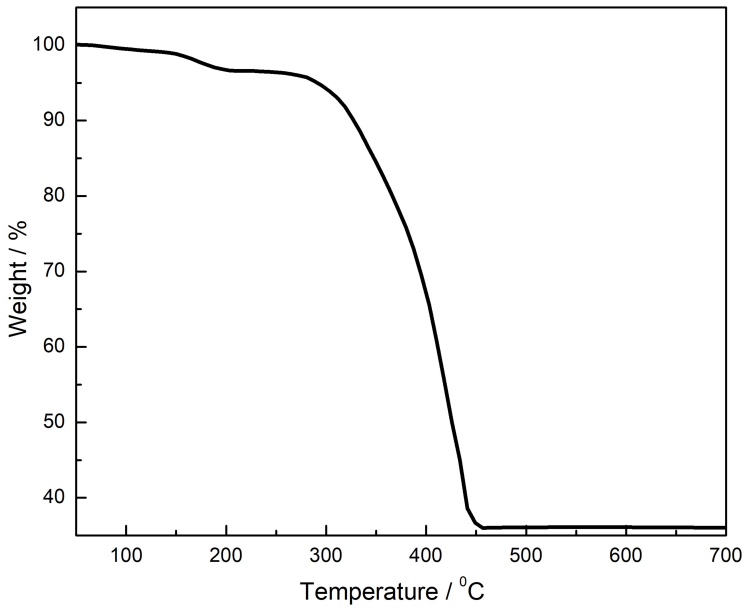


Figure 4. The TGA of the BMMOF11.

improves dispersion compared to the original bulk form. The reduced particle size offers several advantages, such as increased surface area, which promotes better interaction between BMMOF11 and the electrolyte in applications like water splitting. This enhanced geometrical surface area facilitates electrolyte penetration and improves the efficiency of electrochemical reactions. Moreover, the smaller particle size provides more active sites on the BMMOF11 material, potentially enhancing its electrocatalytic performance, particularly for water splitting.[38] But the BET surface area (591 m^2/g) of the BMMOF11 after grinding remains the same as evident from the Figure S4. Energy-dispersive X-ray spectroscopy (EDS) was utilized to perform elemental composition analysis on BMMOF11 both before and after crushing the sample, as shown in Figures 5j and 5k. The primary elements found in the BMMOF11 structure were identified as Co, Cd, C, and O by the analysis. The presence of Au peaks observed by the EDS analysis can be attributed to the application of a gold coating during the sample preparation process for FESEM and EDS. It is probable that Co and Cd were incorporated into the structure of BMMOF11 structure, whereas C is associated with the organic ligands or linkers, and O is a constituent of the framework. The use of grinding to reduce particle size, along with elemental composition analysis, offers significant contributions to the

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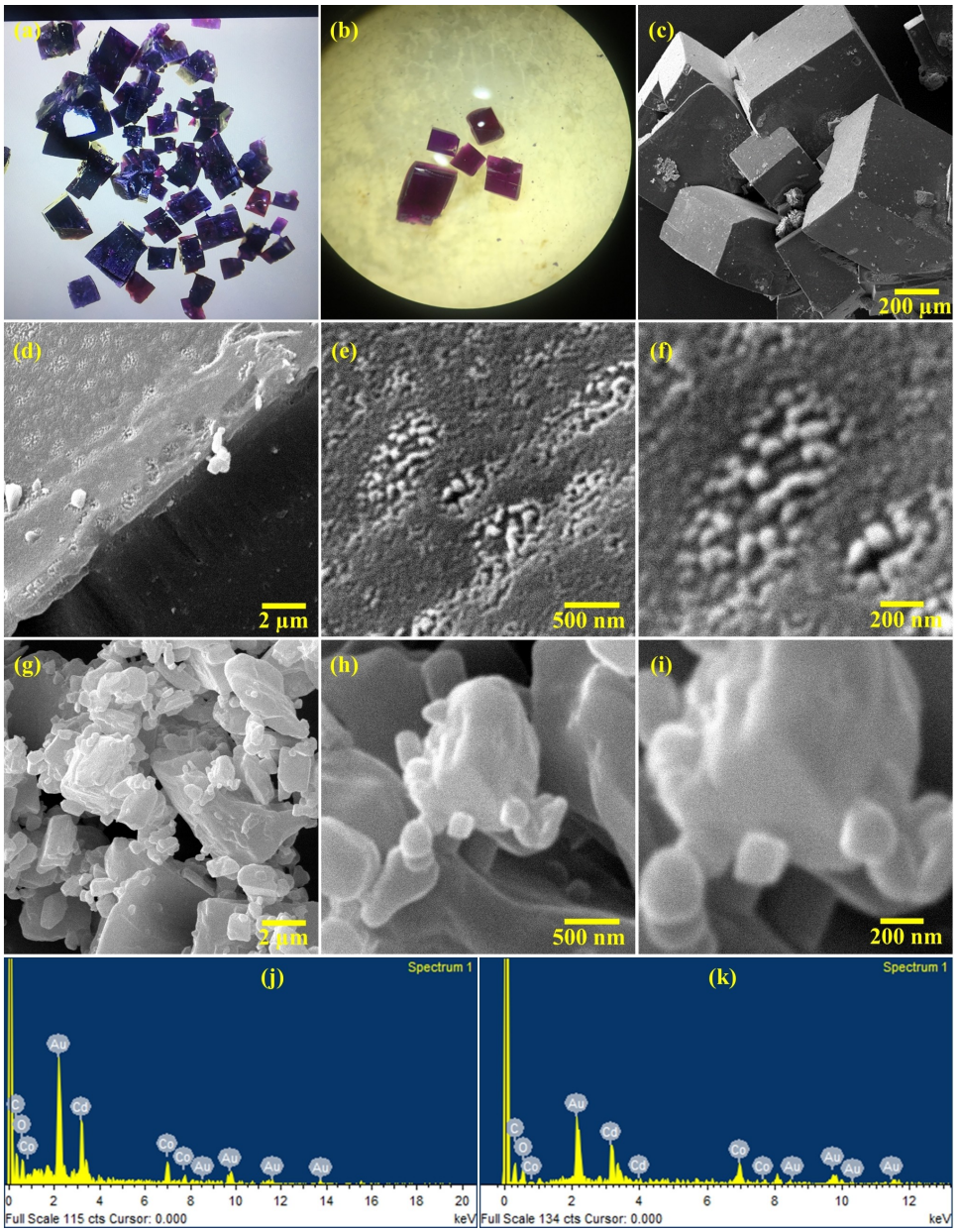


Figure 5. (a,b) Digital photographs of the as-prepared BMMOF11 and different magnification FESEM micrographs of the BMMOF11 (c–f) before and (g–i) after grinding. The corresponding EDS analysis of BMMOF11 (j) before and (k) after grinding.

understanding of the structure and properties of BMMOF11. These findings underscore its promising prospects as an electrocatalyst in several applications, including water splitting. A comprehensive investigation was conducted to analyze the micro/nanostructure of the highly active electrocatalyst, BMMOF11, using state-of-the-art high-resolution transmission electron microscopy (HRTEM). The TEM images (Figures 6a and 6b) evidently revealed the well-dispersed distribution of BMMOF11 particles without any aggregation or clustering. This uniform distribution indicates a well-controlled synthesis process and accessible active sites for efficient catalytic reactions. The HRTEM images (Figures 6c and 6d) provide further evidence for the presence of BMMOF11. These images displayed detailed arrangements and lattice fringes, with an average interlayer spacing of ~0.2 nm, thereby confirming the formation of BMMOF11. The selected area electrode diffraction (SAED) pattern (Figure 6e) supported the crystalline nature of BMMOF11, with distinct diffraction spots indicating a regular atomic arrangement. The line profile analysis of the HRTEM images (Figure 6f) revealed an average interlayer spacing of approximately 0.2 nm, indicating a well-defined and densely packed crystal lattice structure. The consistent interlayer spacing implies the potential for enhanced electrocatalytic activity in the case of BMMOF11.[39,40]

The X-ray photoelectron spectroscopy (XPS) represents critical evidence supporting the composition and electronic state of the BMMOF11 synthesized for water splitting applications. The XPS survey spectrum in Figure 7a provides a broad

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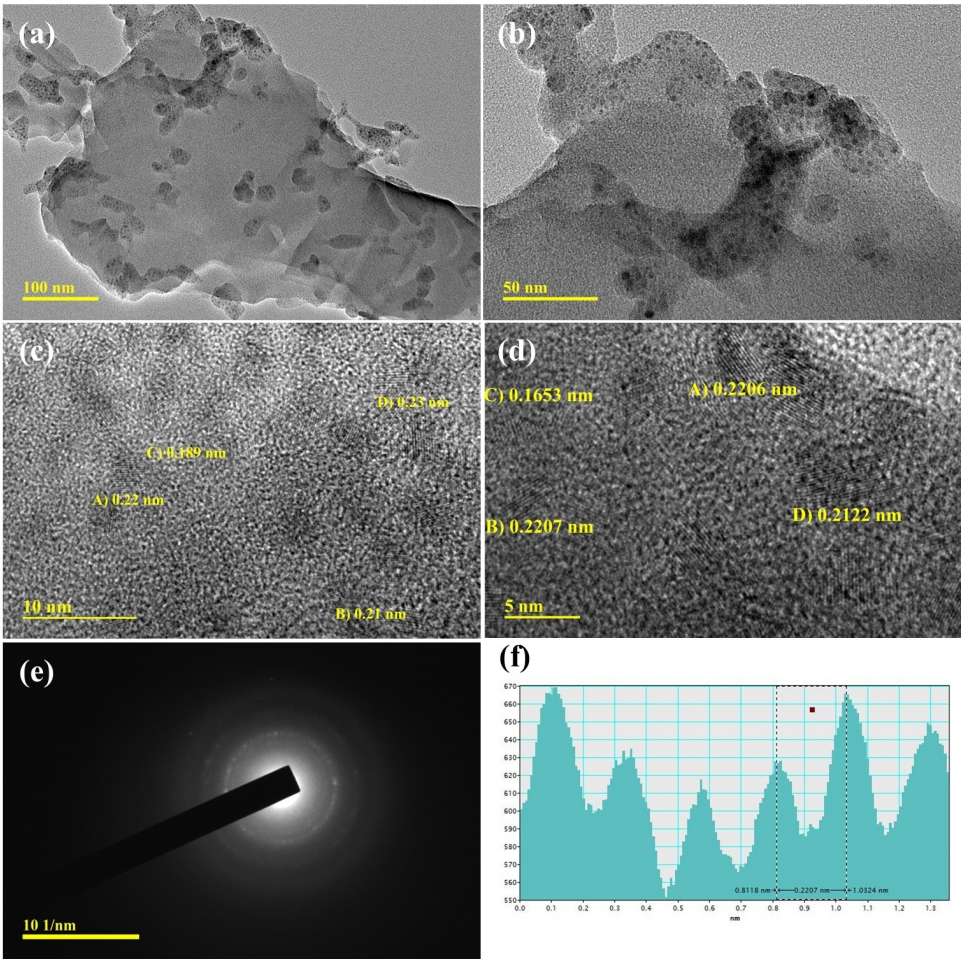


Figure 6. (a,b) TEM images showing the micro/nanostructure of BMMOF11. (c, d) HRTEM images provide detailed insights into the structure of BMMOF11. (e) SAED pattern confirming the crystalline nature of BMMOF11. (f) Lattice spacing profile demonstrating the interlayer spacing within BMMOF11.

overview of the elemental composition. The presence of peaks corresponding to C 1s, Cd 3d, O 1s, and Co 2p at binding energies of approximately 284 eV, 410 eV, 532 eV, and 790 eV, respectively, confirms the presence of these elements in the BMMOF11 structure. These binding energies are consistent with the expected electronic environments for carbon, cadmium, oxygen, and cobalt within the framework, indicating successful incorporation of the targeted metal ions and organic linker (BTC). The high-resolution XPS spectrum of Cd 3d in Figure 7b displays two distinct peaks, characteristic of the spin-orbit coupling in Cd's electronic configuration. The binding energies of around 406 eV for Cd $3d_{5/2}$ and 412 eV for Cd $3d_{3/2}$ are indicative of Cd in a divalent state (Cd^{2+}),[41] which is typical for Cd in a stable MOF environment. The high-resolution XPS spectrum for Co 2p in Figure 7c reveals more detailed analysis, showing two major regions for Co $2p_{3/2}$ and Co $2p_{1/2}$ at around 791 eV and 785 eV, respectively, which are characteristic of Co's electronic states.[42] The peak for Co $2p_{3/2}$ is deconvoluted into three peaks representing Co^{2+}, Co^{3+}, and a satellite peak, suggesting the presence of mixed valence states. Similarly, the peak for Co $2p_{1/2}$ is deconvoluted into two peaks for Co^{2+} and Co^{3+}. This is significant as the presence of multiple oxidation states of cobalt can facilitate redox reactions, which are central to the catalytic activity in electrochemical water splitting. The mixed valence states of Co as indicated by the Co 2p XPS spectrum are particularly important for catalytic purposes. The Co^{2+} and Co^{3+} ions can act as active sites for the water oxidation reaction, where the conversion between the two states during the catalytic cycle can enhance electron transfer processes, leading to improved catalytic efficiency. Based on the XPS analysis, the BMMOF11 demonstrates a suitable elemental composition and the presence of catalytically active sites necessary for water splitting.[43] The MOF structure, with its Co and Cd centers, provides the electronic and structural characteristics that can facilitate the absorption and activation of water molecules, followed by their subsequent dissociation into oxygen and hydrogen. Furthermore, the stability of the BMMOF11 composite under the electrochemical conditions needed for water oxidation, as indicated by the consistent and characteristic XPS peaks, is promising for its application as an electrocatalyst in renewable energy technologies.[44] The XPS spectra not only confirm the successful synthesis of the BMMOF11 with the intended bimetallic composition and coordination environment but also reveal the presence of

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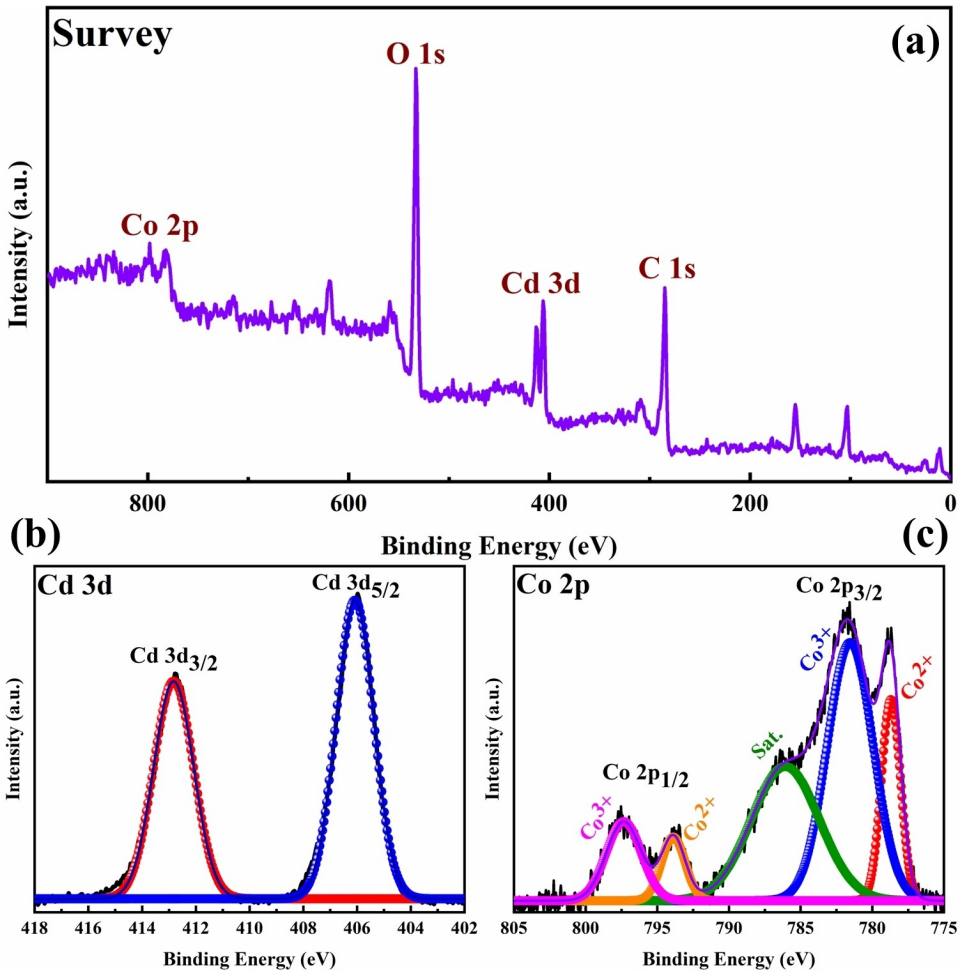


Figure 7. (a) XPS survey spectrum and high resolution XPS spectra of (b) Cd 3d and (c) Co 2p of BMMOF11.

electronic states that are conducive to its function as an effective and stable electrocatalyst for water splitting. This underscores the potential of BMMOF11 as a high-activity, high-stability candidate for large-scale electrochemical applications.

2.3. Electrocatalytic activity for electrochemical water oxidation

The metallic ratio in MOFs could play a significant role in catalytic activities,[45,46] and thus, we performed a comparative study of BMMOF11 with different metal ratios. The cyclic voltammetry (CV) curves depicted in Figure S5 reflect the influence of metal ratios on the OER performance of the synthesized bimetallic BMMOF11 MOFs with varying Co and Cd ratios. The electrocatalytic activity of the different MOFs is benchmarked against the performance of a bare NF and a NF modified with only the organic linker (Linker/NF). BMMOF11-1/NF with a Co:Cd ratio of 1:2 shows a higher current density than the Linker/NF and bare NF, indicating that the presence of bimetal species enhances OER activity. BMMOF11/NF with an equimolar ratio of Co:Cd (1:1) demonstrates the highest current density, suggesting optimal synergistic interaction between Co and Cd at this ratio. BMMOF11-2/NF with a Co:Cd ratio of 2:1 also exhibits high current densities, comparable to BMMOF11/NF, indicating that a higher proportion of Co relative to Cd still maintains high catalytic activity. The varying performance levels can be attributed to the different local environments and electronic interactions between the metal ions, which alter the adsorption energy of OER intermediates and the ease of electron transfer processes. Specifically, the 1:1 Co:Cd ratio seems to provide a balanced electronic structure that facilitates the OER process more efficiently than other ratios. This experimental data aligns with previous studies suggesting that the metal species and their ratios are crucial for OER performance. The highest activity displayed by BMMOF11/NF (1:1 ratio) could be due to a more favorable electronic structure for facilitating OER, which might involve optimal metal-oxygen bond strengths that ease the formation and release of oxygen. The electrochemical water oxidation activities of the BMMOF11 electrocatalyst were further assessed by CV analysis using NF as a substrate in 1 M KOH at a scan rate of 50 mV/sec. As shown in Figure 8a, the BMMOF11/NF electrocatalyst has exceptional activity, with a substantially lower overpotential (0.4 V) required to achieve a current density of 10 mA/cm^2, which is a standard value in evaluating electrochemical water oxidation catalysts.[47]

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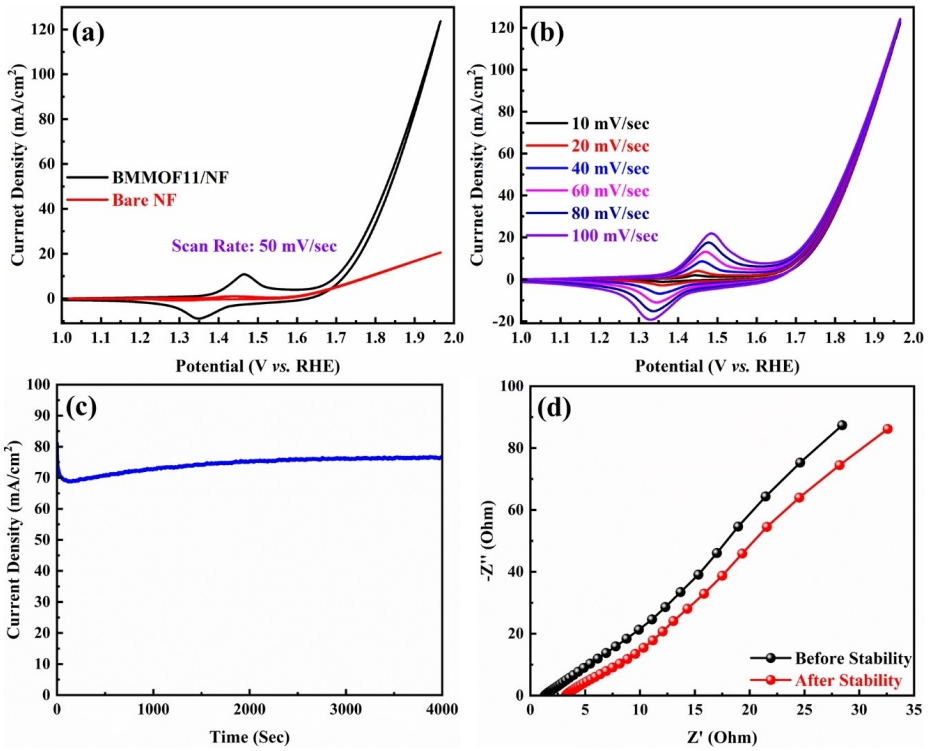


Figure 8. (a) Comparison of CV curves for BMMOF11/NF and bare NF electrodes in 1 M KOH aqueous electrolyte, measured at the same scan rate of 50 mV/sec. (b) CV curves of BMMOF11/NF electrode at different scan rates, (c) amperometric analysis (stability) of BMMOF11/NF electrode in 1 M KOH at 1.9 V *vs.* RHE, and (d) Nyquist plots of BMMOF11/NF electrode before and after stability performance.

Similarly, the electrochemical water oxidation on the BMMOF11/NF electrocatalyst exhibits a very low onset potential (~1.62 V *vs.* RHE). Also, at a high potential (1.9 V *vs.* RHE), the BMMOF11/NF electrocatalyst achieves a current density of about 124 mA/cm^2, which is significantly greater than the bare NF (Figure 8a) and the already reported MOFs-based electrocatalysts.[48–50] The CV curves were then recorded at different scan rates, from 10 to 100 mV/sec (Figure 8b). Interestingly, across all the various applied scan rates (from 10 to 100 mV/sec), no discernible disparities were seen in the initial potential, overpotential, and maximum current density at 1.9 V (*vs.* RHE). Additionally, it is worth noting that achieving a current density of 100 mA/cm^2 with a minimal overpotential of 0.6 V is unusual for NF-supported electrocatalyst systems. The prepared electrocatalyst (BMMOF11/NF) has clear superiority in terms of achieving a high current density while maintaining a low overpotential. The CV curves exhibit pre-peaks, specifically redox peaks, within the voltage range of 1.3 V to 1.6 V (*vs.* RHE). These pre-peaks correspond to the electrochemical processes of oxidation and reduction involving the BMMOF11/NF electrocatalyst.

The stability of the BMMOF11/NF electrocatalyst was assessed by subjecting it to a 1 M KOH aqueous solution at a potential of 1.9 V (*vs.* RHE) for a duration of 4000 sec. Figure 8c illustrates the stability profile of the sample, demonstrating the robustness of the electrocatalyst in facilitating electrochemical water oxidation. The results of the extended electrolysis stability test employing the BMMOF11/NF electrocatalyst indicate its potential for sustained water splitting and oxygen generation. The electrocatalyst demonstrated the ability to operate at a current density of around 70 mA/cm^2, while maintaining an applied voltage of 1.9 V *vs.* RHE. The stability profile demonstrates that the electrocatalyst exhibits remarkable stability even at elevated potentials, hence highlighting the effectiveness of the BMMOF11/NF electrocatalyst in facilitating the process of electrochemical water oxidation. The crystallinity and the morphology of the BMMOF11 was retained after the stability test (Figure S6, and S7). The pore volume also remains unchanged on grinding (Figure S8). The electrochemical performance was evaluated using electrochemical impedance spectroscopy, as shown in Figure 8d. The Nyquist plot of the BMMOF11/NF electrocatalyst exhibited little deviation before and after stability performance.

Also, the BMMOF11/NF electrocatalyst exhibited a minimal solution resistance of ~1.3 Ω, which was increased to ~2.3 Ω after stability. The minimal existence of semicircles and good straight-line slope at low-frequency regions in the Nyquist plots also show very little charge transfer resistance and good diffusion of electrolyte ions in the pores of the BMMOF11/NF electrocatalyst, which are beneficial for excellent electrochemical performance. By considering the above-mentioned parameters, the BMMOF11/NF electrocatalyst could be considered a superior electrocatalyst for electrochemical water oxidation. These findings demonstrate that the BMMOF11 can support water oxidation for various real-world uses, including producing

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pure H_2 and O_2 and converting CO_2 into valuable renewable fuel.

Using the CV analysis in the non-Faradaic zone, the electrochemical surface areas (ECSA) of the BMMOF11/NF electrocatalyst were calculated from its double-layered capacitance (C_{dl}).[51] The charging of the double layer is the source of the current that is generated in the non-Faradaic zone, and it was found to have a linear connection with the amount of active surface area. The CV curves were obtained by conducting experiments within the potential range of 0.97 V to 1.16 V (*vs.* RHE) in a 1 M KOH aqueous solution. The scan speeds employed ranged from 10 to 100 mV/sec, as illustrated in Figure 9a. It has been reported that 1 cm^2 of flat surface area has a specific capacitance (C_s) value that ranges from 0.02 to 0.06 mF/cm^2, with an average value of 0.04 mF/cm^2.[52–54] The following equation (1) has been used to convert the C_{dl} value to the ECSA.[12]

$$ECSA = \frac{C_{dl}}{C_s} \quad (1)$$

In Figure 8b, the slope of the straight line represents C_{dl} which is equal to 4.29 mF. Using equation (1) and the aforementioned C_{dl} and C_s values, the ECSA was calculated for the BMMOF11/NF electrocatalyst, which was ~107 cm^2. Further, the electrochemical activity evaluation of the BMMOF11/NF electrocatalyst was examined by a Tafel plot. As can be observed from Figure 9c, the BMMOF11/NF electrocatalyst exhibits a higher electropositive corrosion potential (Ecorr) of ~1.09 V *vs.* RHE, and the corrosion current density (Icorr) of the BMMOF11/NF electrocatalyst is ~86 μA/cm^2, which shows a high corrosion resistance.[55–57] Furthermore, the BMMOF11/NF electrocatalyst demonstrated an anodic Tafel slope (β_a) of ~85 mV/decade and a cathodic Tafel slope (β_c) of ~135 mV/decade, which further confirms that the BMMOF11/NF is an excellent electrocatalyst for water oxidation.[55,57,58] Table 1 presents a comprehensive overview of different MOF-based electrocatalysts for the OER. The results indicate that the prepared BMMOF11 MOF exhibits superior or comparable performance compared to the majority of MOFs utilized as electrolytes for OER.

In this study, we also investigated the catalytic performance of the BMMOF11 electrocatalyst through turnover frequency (TOF) analysis using CV at varying scan rates within an

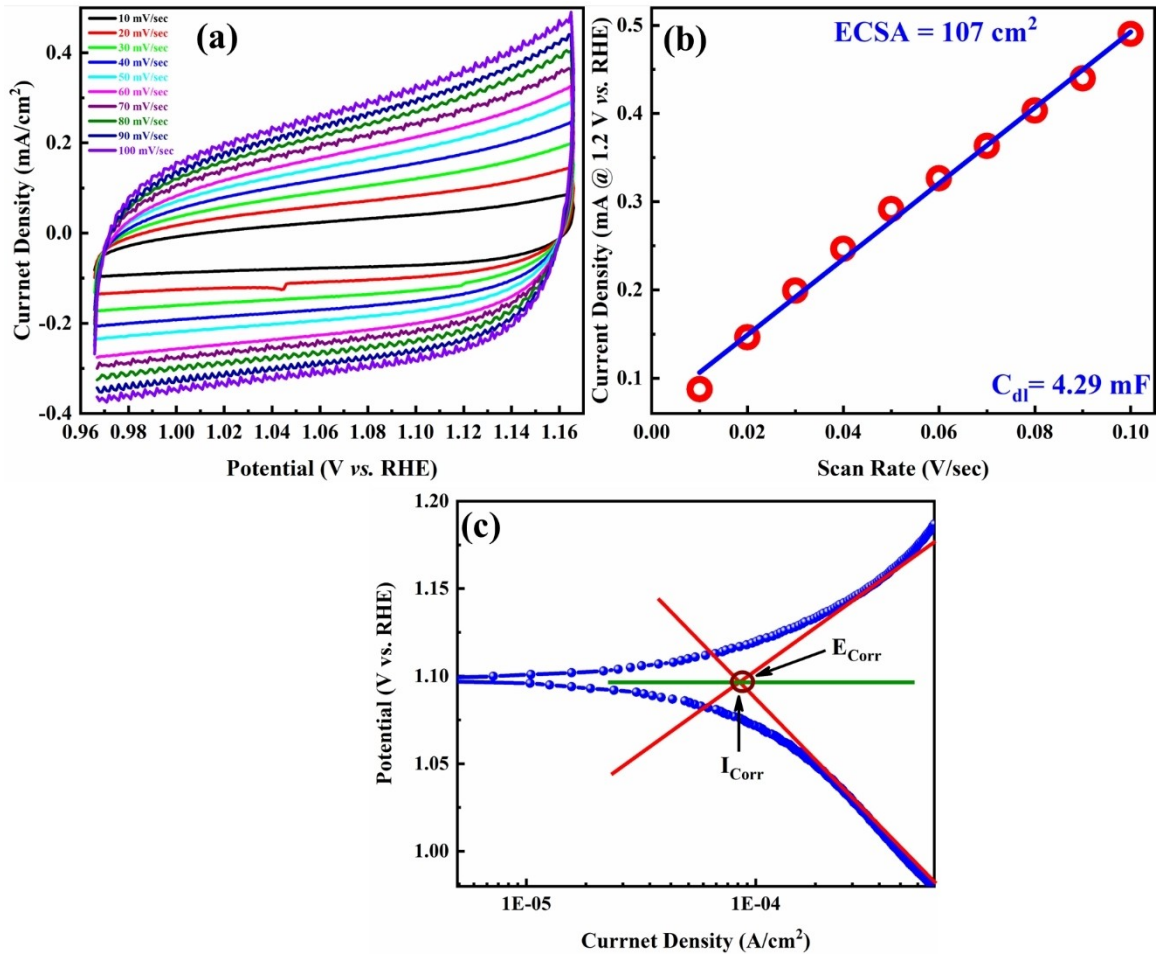


Figure 9. (a) CV curves of the BMMOF11/NF electrode in 1 M KOH aqueous electrolyte at different scan rates (10, 20, 30, 40, 50, 60, 70, 80, 90, and 100 mV/sec), (b) the corresponding anodic charging current of the CV measurements, measured at 1.2 V *vs.* RHE, plotted as a function of the scan rate, and (c) Tafel polarization plot for the BMMOF11/NF electrocatalyst.

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Table 1. A summary of various MOF-based electrocatalysts for OER reported in the literature.

Electrocatalysts	Electrolyte	Overpotential η [mV] at 10 mA/cm^2	Tafel Slope [mV/decade]	TOF (sec^{-1}) @ Overpotential η [mV]	Ref.
BMMOF11	1 M KOH	0.4 V	85	0.04@300 0.32@735	**This Work**
Fe3–BPTC	0.1 M KOH	0.365	81.8	0.002@400	[61]
POM@ZIF8	0.1 M Na_2SO_4	0. 784	783.62	…	[62]
CUMSs–ZIF-67	1 M KOH	0.41	185.1	0.04@300	[63]
Fe_2Co–BPTC	1 M KOH	0.476	87.2	…	[64]
CeO_2/MOF–GO	1 M KOH	0.386	98.1	…	[65]
Co–BPDC/Co–BDC	1 M KOH	0.335	81.7	…	[66]
Co–MONs	1 M KOH	0.309	75.71	0.03@300	[67]
Co–N–CNTs	0.1 M KOH	0.46	…	…	[68]
Co–BDC	1 M KOH	0.378	78	0.03@250	[69]

overpotential window of 300 to 735 mV. The resulting TOF values, ranging from 0.04 to 0.32 sec^{-1}, revealed a compelling trend: an increase in the scan rate from 10 to 100 mV/sec corresponded to a consistent elevation in TOF, as shown in Figure 10. The TOF exhibited by the BMMOF11 electrocatalyst demonstrates comparable or greater performance in terms of TOF when compared to a similar electrocatalyst based on MOF chemistry, as illustrated in Table 1. The observed behavior highlights the effectiveness of the BMMOF11 electrocatalyst in promoting the process of OER, indicating enhanced kinetics and mass transport.[59,60] Consequently, the catalyst demonstrates exceptional potential for applications requiring efficient oxygen evolution, such as water electrolysis and renewable energy technologies, making it a promising candidate for advancing electrochemical systems aimed at sustainable energy production and storage.

The plausible electrocatalytic oxidation of water on the bimetallic BMMOF11/NF MOF as an electrocatalyst could be explained as follows: The first step in the water splitting reaction is the adsorption of water molecules on the surface of the BMMOF11 electrocatalyst. The metal centers (Co and Cd) in BMMOF11 have active sites that can coordinate with water molecules. The water molecules approach the catalyst surface and bind to the metal centers through oxygen coordination.[70]

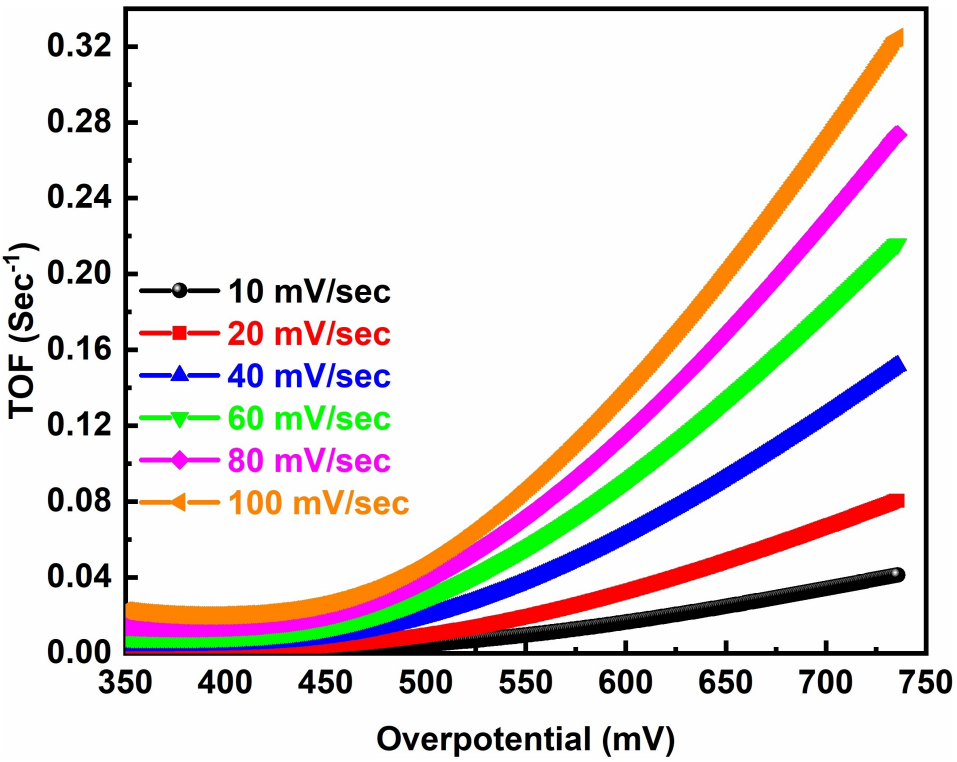


Figure 10. TOF trends in OER as a function of overpotential and scan rate.

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Once the water molecules are adsorbed, a proton-coupled electron transfer (PCET) process occurs.[71] The PCET involves the transfer of both a proton and an electron. In this step, one water molecule donates a proton (H^+) to the catalyst surface, while another water molecule donates an electron. After the PCET process, the donated proton (H^+) combines with another water molecule, forming a hydroxyl group (OH^-) on the catalyst surface. Meanwhile, the electron transfer from the water molecule reacts with a metal center in BMMOF11, reducing the metal center. In the OER, the hydroxyl group (OH^-) formed in the previous step is further oxidized to produce oxygen gas (O_2). The hydroxyl group undergoes a series of oxidation steps involving the transfer of electrons to the catalyst and the release of oxygen atoms. These oxygen atoms combine to form O_2 molecules. The overall reaction for water splitting using BMMOF11 as the electrocatalyst can be represented as follows in equation (2):[71]

$$2H_2O \rightarrow O_2 + 4H^+ + 4e^- \quad (2)$$

Equation (2) shows that at two water molecules are split into oxygen gas (O_2) and four protons (H^+), along with four electrons (e^-), as shown in Figure 11a. The proposed mechanism provides a general understanding of the critical steps of the electrocatalytic water splitting process using BMMOF11.

The BMMOF11/NF electrocatalyst, synthesized using Co and Cd in a MOF, exhibits exceptional electrocatalytic activity for OER. The CV analysis and Tafel plot provide crucial insights into the underlying OER mechanism. Given the findings, the Tafel slope values observed for BMMOF11/NF are approximately 85 mV/decade (anodic) and 135 mV/decade (cathodic). These values are key indicators of the OER mechanism. Typically, a lower Tafel slope suggests a more efficient electrocatalytic process, indicating fewer steps or a faster rate-determining step in the overall reaction. In the context of Xie et al. mechanisms,[72] a lower Tafel slope is associated with the lattice oxygen mechanism (LOM), while higher Tafel slopes correspond to the adsorbate evolution mechanism (AEM). The conventional AEM and LOM mechanisms, where M represents the active sites, are shown in Figure 11b and Figure 11c, respectively.[72] Given the Tafel slopes for BMMOF11/NF, it appears that the OER process may involve a mixed mechanism or a slightly varied pathway from the conventional AEM. The presence of both Co and Cd and their ratios seems to play a significant role in this behavior. Co and Cd may facilitate different steps in the OER process, possibly influencing both the adsorption of intermediates and the involvement of lattice oxygen. Furthermore, a significant ECSA in BMMOF11/NF, along with a low onset potential for water oxidation and high current density, suggests a highly efficient electrocatalytic process.[73] This efficiency could be due to the optimal Co:Cd ratio, which provides a balanced electronic structure, potentially facilitating both AEM and LOM pathways in the OER. The electrocatalytic OER mechanism in BMMOF11/NF, as evidenced by the Tafel analysis, seems to be influenced by a combination of factors including the unique electronic interactions between Co and Cd, the adsorption energy of OER intermediates, and potentially the involvement of lattice oxygen. The exact pathway may be a hybrid of AEM and LOM, with the precise contribution of each mechanism requiring further investigation, possibly through more detailed spectroscopic analysis or computational modeling. This nuanced understanding of the OER mechanism in BMMOF11/NF underscores its potential as a highly efficient and novel electrocatalyst for OER applications.

One of the key advantages of BMMOF11/NF as an electrocatalyst is its ability to lower the activation energy and overpotential required for the water splitting reaction. The presence of the catalyst provides an alternative reaction path-

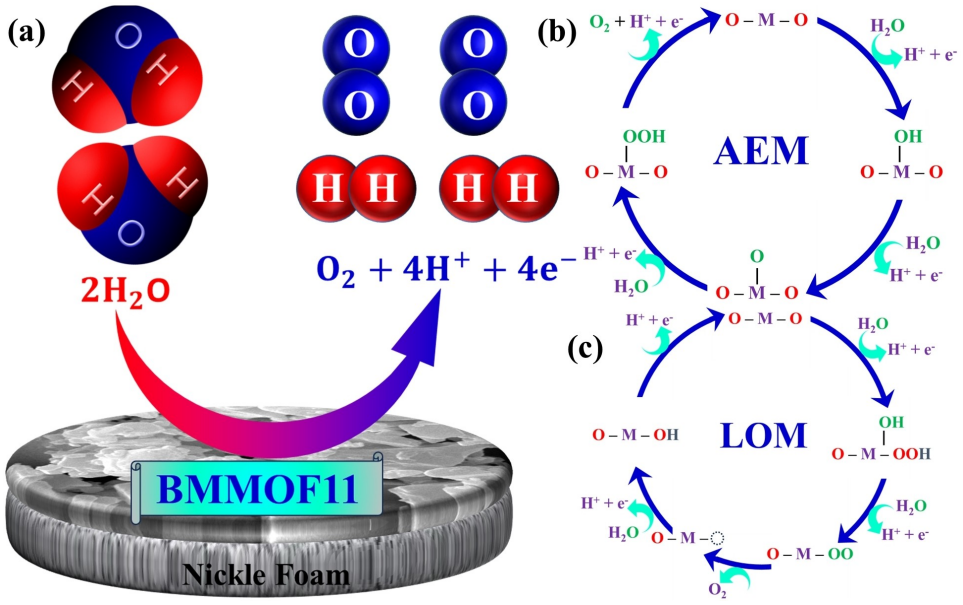


Figure 11. (a) Schematic representation of the electrocatalytic water splitting process utilizing BMMOF11/NF as the catalyst. Conventional (b) AEM and (b) LOM mechanisms where M represents the active sites.

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way with lower energy barriers, allowing the reaction to proceed more efficiently and at lower applied voltages. BMMOF11/NF exhibits excellent catalytic activity for the OER, which produces oxygen gas (O_2) during water splitting. The catalyst enhances the kinetics of the OER, enabling the efficient and rapid formation of oxygen molecules.

BMMOF11/NF demonstrates good stability and durability during the water splitting process. It maintains its catalytic activity over extended periods of electrolysis, allowing for continuous and long-term water splitting operations without significant performance degradation. This stability is crucial for practical applications of water splitting technology.[74] The ECSA of BMMOF11/NF is also an essential factor in its electrocatalytic performance. The high ECSA of the catalyst provides a larger active surface area, allowing for more efficient water adsorption and increased reaction rates. This leads to higher current densities and improved overall electrocatalytic activity.[75]

Similarly, Co plays a crucial role as an OER catalyst in the water splitting reaction. During the OER, Co undergoes oxidation, promoting the formation of oxygen gas (O_2). Co acts as a redox-active center, facilitating the transfer of electrons and promoting the breaking of water molecules to generate oxygen. Cd serves as both a promoter and a structural component in the BMMOF11/NF electrocatalyst. Cd ions contribute to the overall stability and structure of the bimetallic MOF. The presence of Cd helps enhance the stability and durability of the catalyst during the water splitting process. The incorporation of Co and Cd into the BMMOF11/NF electrocatalyst leads to a composite effect, enhancing the overall catalytic activity due to the combined contributions of varying compositions of Co and Cd. The interaction between Co and Cd ions within the MOF framework creates a unique environment that promotes efficient electron transfer and facilitates the water oxidation reaction. Both Co and Cd ions exhibit redox activity, allowing for the transfer of electrons during the water splitting process. The redox activity of these metal ions enables the exchange of charges between the catalyst and water molecules, facilitating the desired redox reactions involved in water splitting. The presence of Co and Cd in the BMMOF11/NF electrocatalyst also allows for tuning its electronic properties. The catalytic activity and selectivity of the catalyst are influenced by the unique electronic configuration and coordination environment of Co and Cd ions. This tuning of electronic properties contributes to the efficient water splitting performance of BMMOF11/NF.

Besides, KOH is a strong electrolyte that dissociates into potassium ions (K^+) and hydroxide ions (OH^-) in solution. These ions facilitate the conduction of ionic conductivity within the electrolyte. A conductive electrolyte is crucial for the movement of ions and charge transfer during water splitting. In the OER, hydroxide ions (OH^-) participate in the electrochemical reaction by accepting electrons and forming oxygen gas (O_2). The KOH electrolyte supplies the OH^- ions, allowing for the continuous regeneration of OH^- ions during the water splitting process.[76] The KOH electrolyte helps maintain a stable and appropriate pH environment for efficient water splitting. A slightly alkaline pH (around 14) is favorable for OER and facilitates the desired electrochemical reactions. The presence of KOH helps to maintain this alkaline pH in the electrolyte solution. As an alkaline electrolyte, KOH reduces the overpotential required for the electrochemical reactions involved in water splitting. A lower overpotential means less energy is needed to drive the desired reactions, improving the overall energy efficiency of the electrolysis process. The KOH electrolyte serves as a medium for the transport of ions between the electrodes. It allows the movement of potassium ions (K^+) and hydroxide ions (OH^-) from one electrode to another, facilitating the completion of the water splitting reaction. Finally, BMMOF11/NF demonstrates notable efficacy as an electrocatalyst for the process of water splitting. This is achieved through the provision of active sites that facilitate the adsorption of water molecules, thereby enabling PCET processes. Additionally, BMMOF11/NF effectively reduces the activation energy and overpotential required for water splitting. Furthermore, it acts as a catalyst for the OER, exhibits remarkable stability, and possesses a substantial electrochemical surface area. These features make BMMOF11/NF a promising candidate for large-scale water splitting and other electrochemical applications aiming to produce clean hydrogen and oxygen or convert CO_2 into renewable fuels.

3. Conclusions

In conclusion, a MOF-based bimetallic electrocatalyst composed of Co and Cd metals (BMMOF11) was successfully synthesized. The innovative zig-zag chains of alternate and vertex-sharing CdO and CoO contorted octahedra bridged by the BTC linker resulted in the 3D network that confirmed its coherence and connectivity. The characterization techniques demonstrated its crystallinity, microporous nature, thermal stability, and the potential for enhanced electrocatalytic activity through particle size reduction. The BMMOF11 electrocatalyst was immobilized on NF for electrochemical measurements and showed excellent electrocatalytic properties towards electrochemical water oxidation in a 1 M KOH aqueous solution. The resulting BMMOF11/NF electrocatalyst delivered excellent electrochemical performance by producing a high current density of ~124 mA/cm^2 at 1.9 V *vs.* RHE and a lower overpotential. The BMMOF11/NF electrocatalyst also showed good stability during electrochemical water oxidation at high potential and high current density. The BMMOF11/NF electrocatalyst has a remarkable level of efficiency, making it a promising candidate for many electrochemical applications on a wide scale. These applications encompass the generation of clean H_2 and O_2 through water splitting, as well as the conversion of CO_2 into valuable fuels.

Conflict of Interests

The authors declare no conflict of interest.

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Keywords: bimetallic electrocatalysts · hydrogen evolution reaction · MOFs · oxygen evaluation reaction · water splitting

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